

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

**COURSE NAME: COMPUTER VISION
ITA0510**

COURSE CODE:

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage:

50%

QUESTIONS: 5

Total Marks50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

*I **AKHILA POCHIMIREDDY (192425225)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: P.Akhila

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	
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1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

Answer:

Grey level transformation is used to improve the visual quality of an image by modifying the intensity values of pixels. It helps to:

- Enhance image contrast.
- Highlight important features.
- Brighten or darken images.
- Improve visibility for further image processing.

(b) Apply a negative transformation $s=255-r$ to the pixel values:
 $r=[50,100,150,200]$

Compute the transformed values.

Answer:

Formula:

$$s=255-r$$

Given:

$$r=[50,100,150,200]$$

Calculation:

- $255 - 50 = 205$
- $255 - 100 = 155$
- $255 - 150 = 105$
- $255 - 200 = 55$

Answer

[205,155,105,55]

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

Log transformation expands the darker pixel values while compressing brighter pixel values. This makes details in dark regions more visible without significantly affecting bright areas.

(b) Given $s = c \log_{10}(1+r)$, where $c=10$, compute s for:
 $r=[0,10,50,100]$

Given

$$s = c \log(1+r)$$

where

$$c=10$$

Using natural logarithm (ln):

For $r = 0$

$$10 \ln(1) = 10(0) = 0$$

For $r = 10$

$$10 \ln(11) = 10(2.3979) = 23.98$$

For $r = 50$

$$10 \ln(51) = 10(3.9318) = 39.32$$

For $r = 100$

$$10 \ln(101) = 10(4.6151) = 46.15$$

Answer

r	s
0	0
10	23.98
50	39.32
100	46.15

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

Histogram equalization redistributes the pixel intensity values over the entire available range. It improves image contrast, especially when the image has poor brightness distribution.

Benefits:

- Improves visibility.
- Enhances contrast.
- Makes image details clearer.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

Answer:

Total pixels

$$N = 2 + 2 + 4 + 8 = 16$$

Cumulative Frequency

Intensity	Frequency	Cumulative
0	2	2
1	2	4
2	4	8
3	8	16

For a **3-bit image**

$$L = 8$$

Equalization formula

$$s = \frac{CDF}{N} (L - 1)$$

where

$$L - 1 = 7$$

Calculations

Intensity 0

$$\frac{2}{16} \times 7 = 0.875 \approx 1$$

Intensity 1

$$\frac{4}{16} \times 7 = 1.75 \approx 2$$

Intensity 2

$$\frac{8}{16} \times 7 = 3.5 \approx 4$$

Intensity 3

$$\frac{16}{16} \times 7 = 7$$

Answer

Original	CDF	Equalized
0	2	1
1	4	2
2	8	4
3	16	7

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

Smoothing removes noise and reduces small intensity variations by replacing each pixel with the average of its neighboring pixels. It produces a smoother image.

Applications:

- Noise removal
- Image preprocessing
- Blur unwanted details

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

Answer:

Matrix

10	20	30
20	30	40
30	40	50

Sum

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average

$$\frac{270}{9} = 30$$

Answer

30

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

Sharpening emphasizes edges and fine details by increasing the intensity difference between neighboring pixels. It is mainly used for edge enhancement and improving image clarity.

Applications:

- Edge detection
- Medical imaging
- Object recognition

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

to the matrix:

10 10 10, 10 20 10, 10 10 10

Compute the output at the center.

Answer:

Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image

$$\begin{array}{ccc} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{array}$$

Calculation

$$\begin{aligned} &(4 \times 20) - (10 + 10 + 10 + 10) \\ &80 - 40 = 40 \end{aligned}$$

Answer

$$\boxed{40}$$